

# FROM EINSTEIN TO NEWTON

## EINSTEIN'S EQUATION

$$G_{\alpha\beta} + \Lambda g_{\alpha\beta} = 8\pi G \langle T_{\alpha\beta} \rangle$$

weak field, slow motion

## POISSON'S EQUATION

$$\nabla^2 \Phi = 4\pi G \rho$$

point mass

## GRAVITATIONAL POTENTIAL

$$\Phi = -GM/r$$

$$F = -m\nabla\Phi$$

Geodesic eq.

## NEWTON'S LAW

$$F = -GMm/r^2$$

*Each step is a well-defined limit of the previous one*